

LayerRoPE

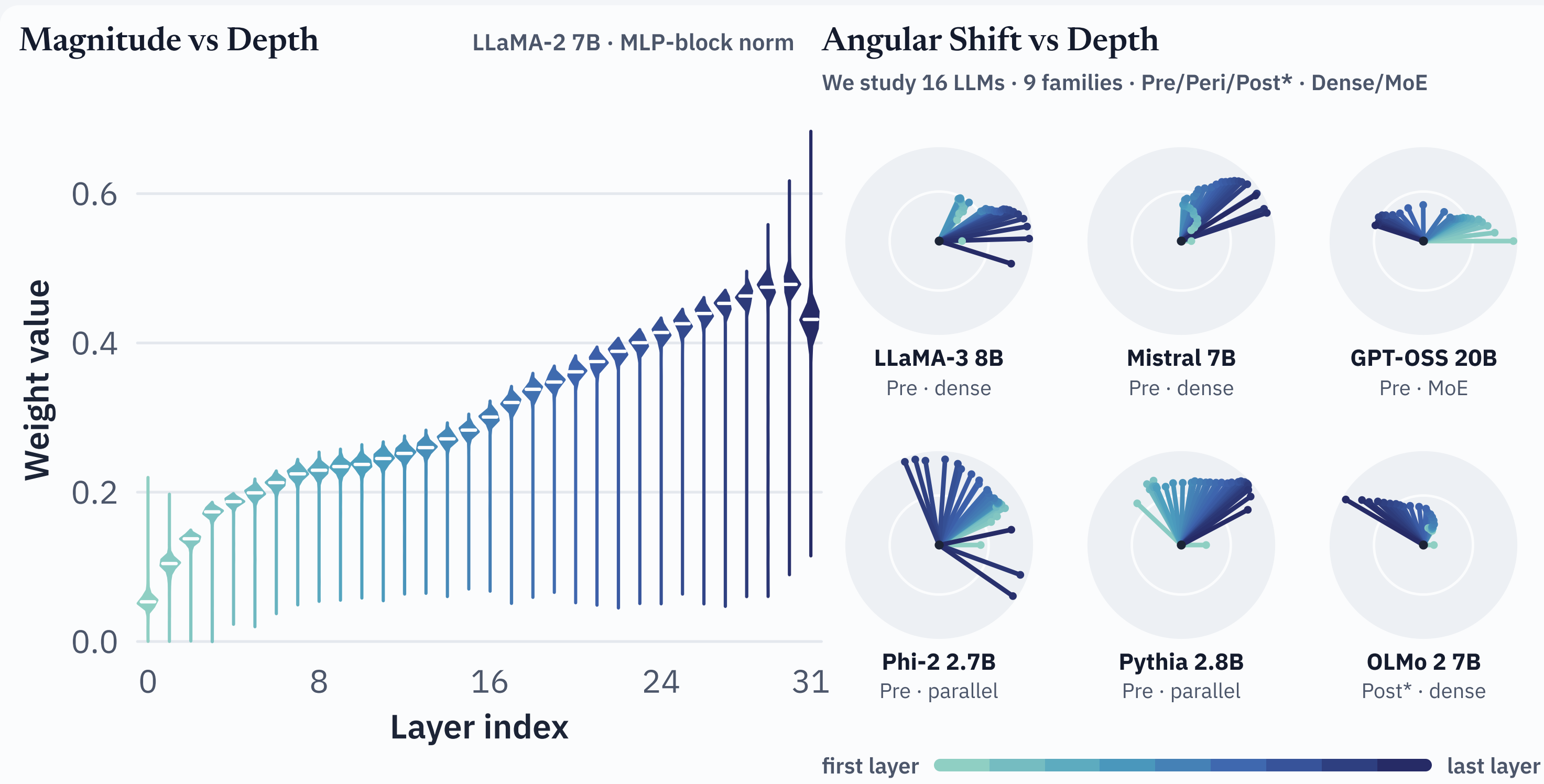
Dynamic Depth-wise Magnitude & Angular Superposition

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TL;DR: Hidden-state norms, variance grow by orders of magnitude with depth, a “curse of depth” treated as a pathology. We find it encodes an **emergent depth code** in the norm weights γ , and make it explicit: **LayerRoPE is like RoPE along depth**, and it learns to amplify variance!

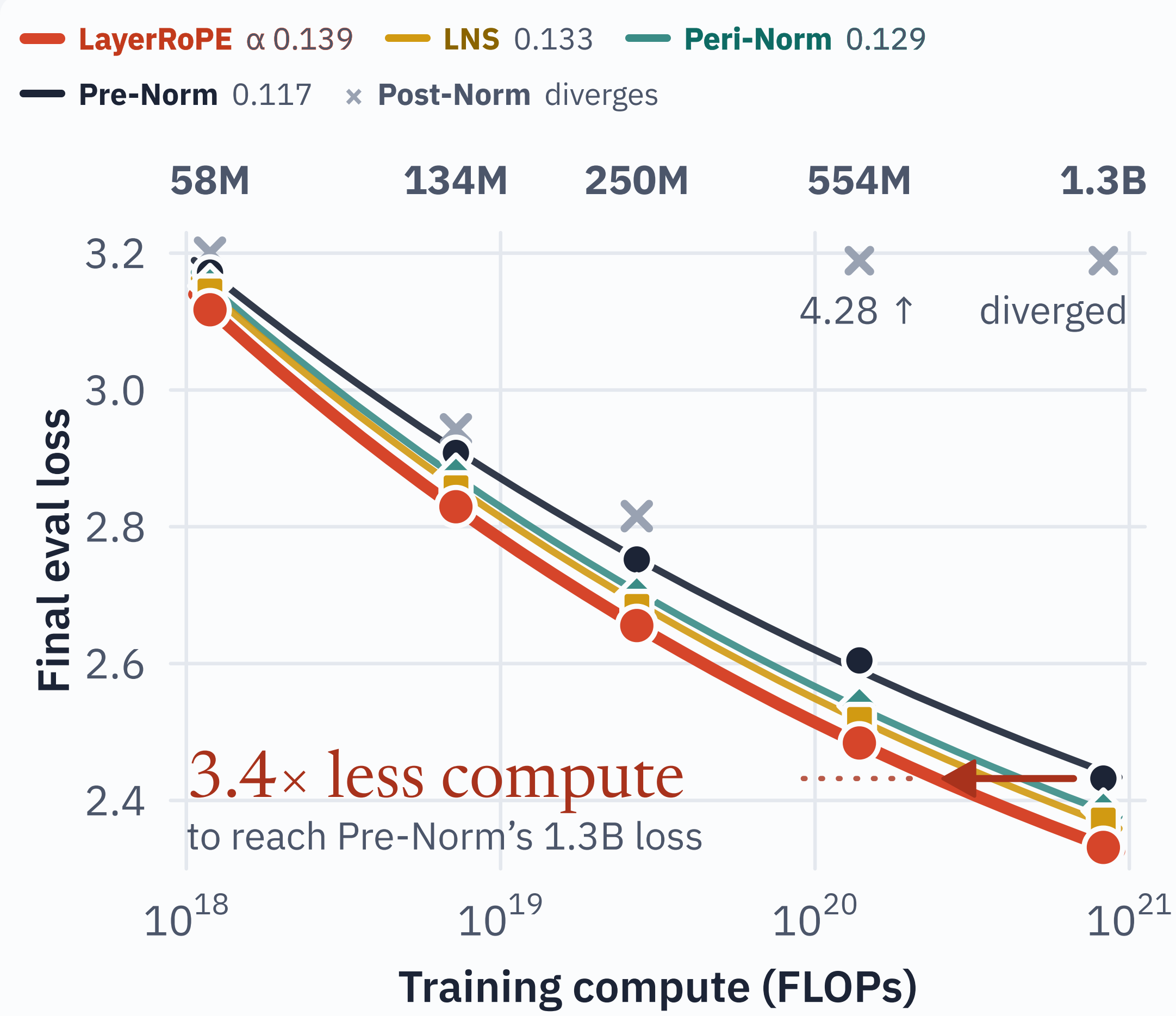
1 Norm weights encode depth

The RMSNorm weight γ is the only learned per-layer gain on the residual stream. In pre-trained LLMs, the MLP-block γ **grows and rotates with depth**, jointly encoding the layer index.



4 Better Compute Scaling

58M-1.3B Model Scaling Ladder. 4× Chinchilla budget (upto 107B tokens)



C4 Pre-Training. Carefully LR Tuned Baselines. Power-law fits with a shared $E = 1.81$. LayerRoPE’s margin over Pre-Norm **roughly doubles** from 58M to 1.3B.

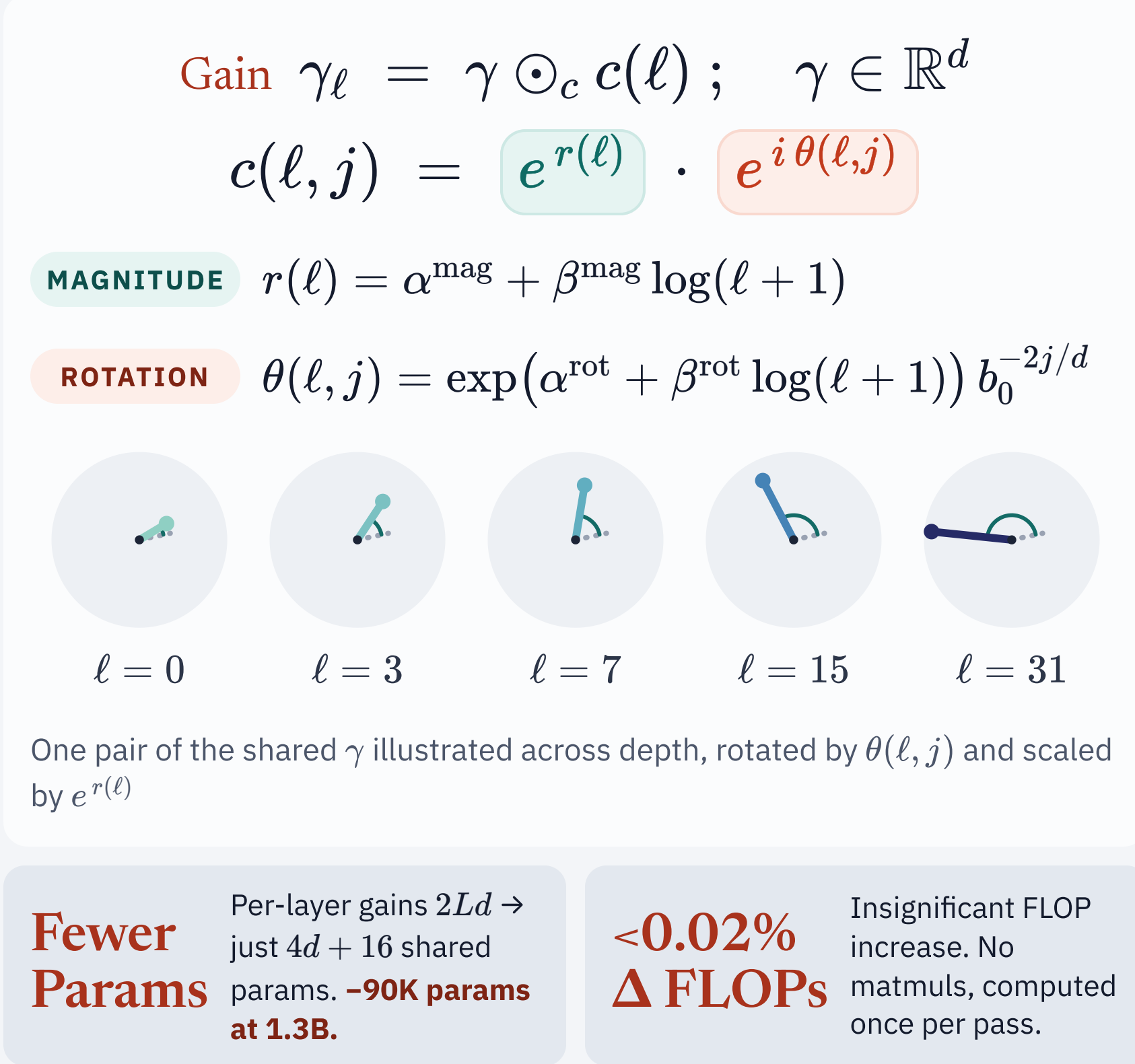
7 Downstream Performance

	1.3B · 107B tokens		6.7B · 11.7B tokens	
	avg acc ↑	LAMBADA ppl ↓	val loss ↓	avg acc ↑
LayerRoPE	52.6	14.0	2.377	49.3
Peri-Norm	51.6	15.7	2.514	46.2
LNS	50.9	16.7	2.464	47.0
Pre-Norm	48.0	21.0	2.545	46.2

Zero-shot avg of 8 tasks · 6.7B at one shared, untuned LR · Post-Norm fails at 1.3B.

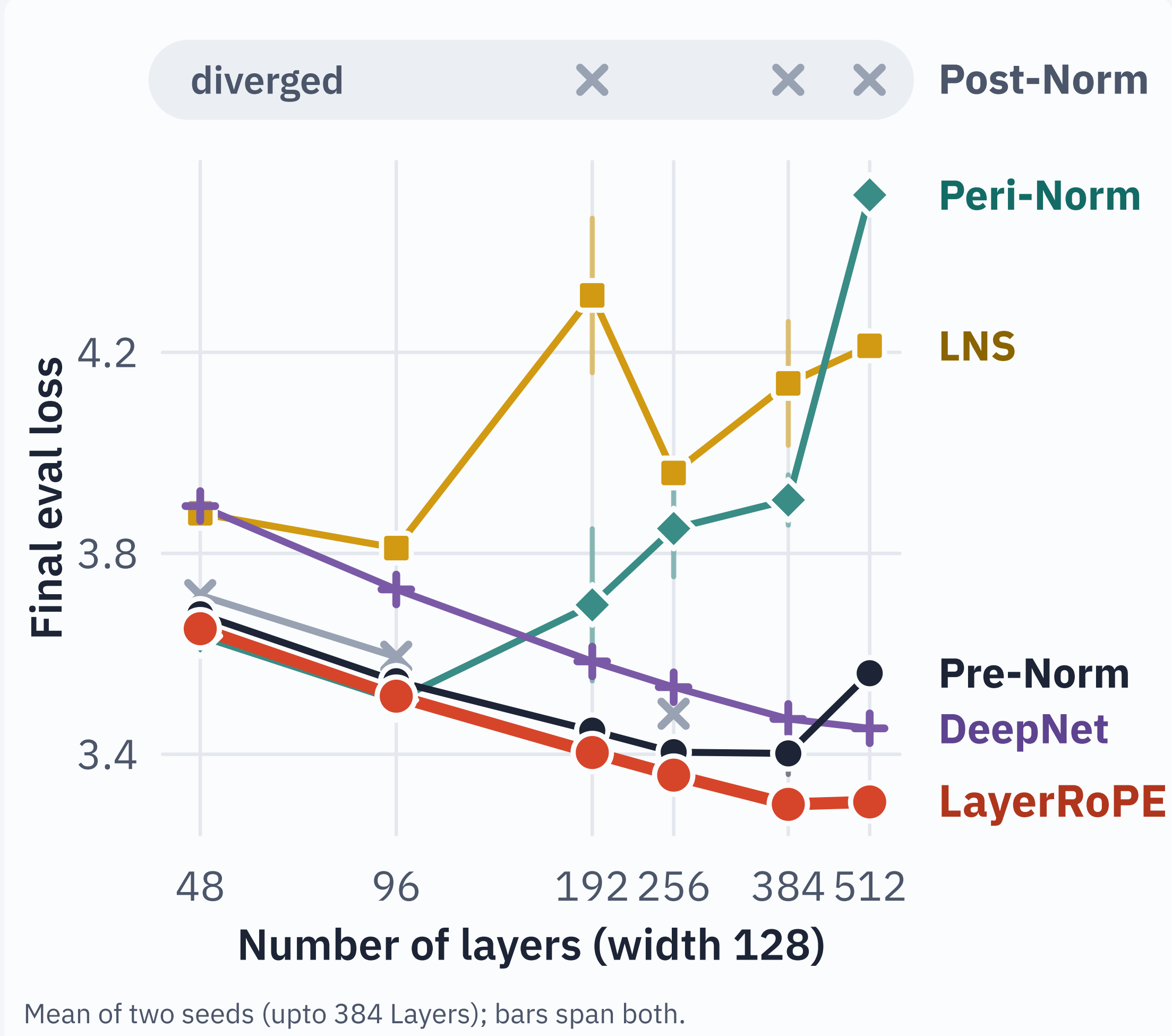
2 LayerRoPE makes Depth-encoding explicit

Analogous to RoPE, but along depth. Applied at read & write of every Attention, MLP block. Shared γ is rotated, scaled by the layer index ℓ to give each per-layer gain γ_ℓ .



5 Depth Scaling to 512 layers

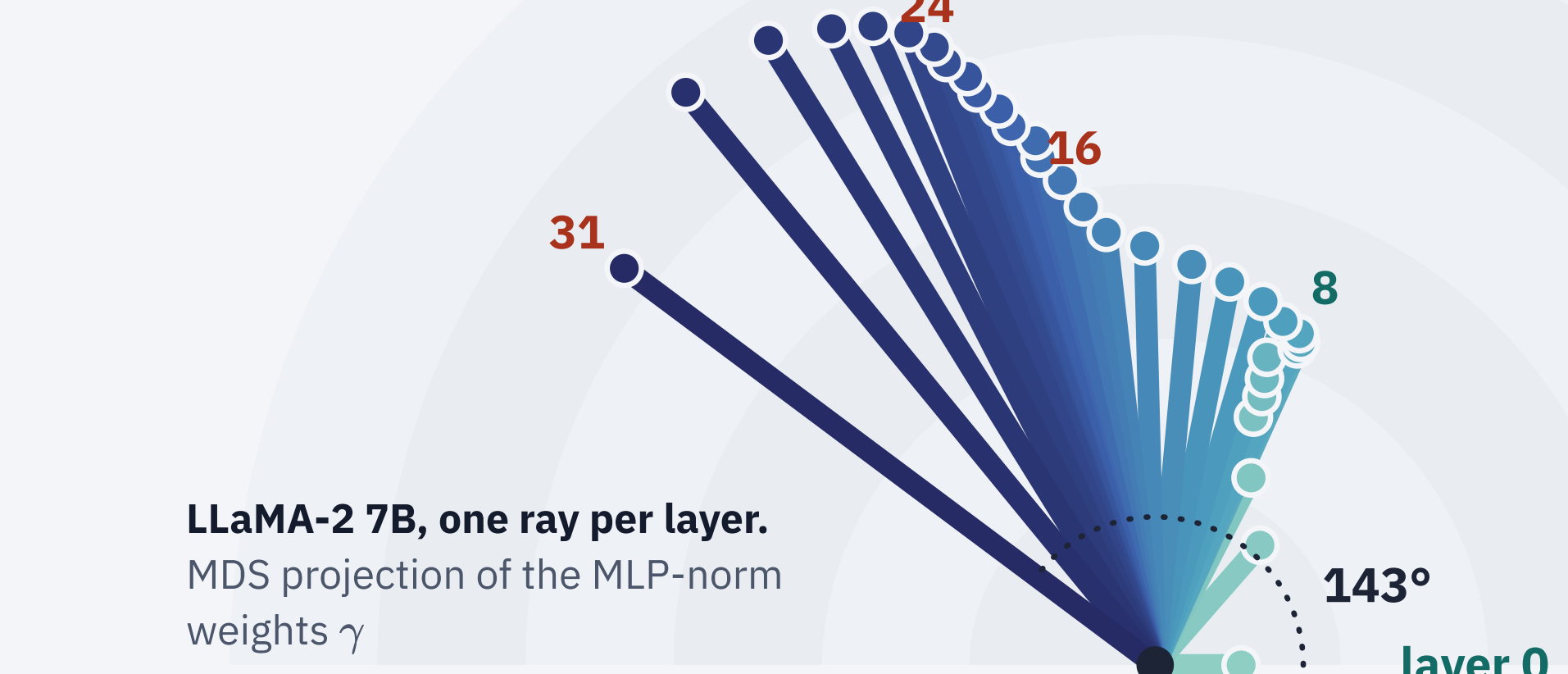
Scaling depth from 48 to 512 layers (Width 128 fixed, 18M–111M params), Shared LR (Tuned LR Depth scaling also similarly better)



8 Vision Transformers & Looped Latent Models

Looped LMs · Parcae	(Default) Parcae $\rightarrow$ Parcae + LayerRoPE · $T=8$ loops	
	val ppl ↓	avg acc ↑
140M	18.66 $\rightarrow$ 18.36	33.0 $\rightarrow$ 34.9
370M	14.22 $\rightarrow$ 14.05	43.4 $\rightarrow$ 44.0
770M	12.49 $\rightarrow$ 12.09	50.1 $\rightarrow$ 50.6

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3 Amplifying Residual Variance Helps ?

Prior methods damp the residual stream. LayerRoPE learns to **scale down reads** and **scale up writes**.

